

MA 560: Nonlinear Dynamics and Chaos

Course Code	MA 560
Course Name	Nonlinear Dynamics and Chaos
L-T-P-C	3-0-0-3
Prerequisites	IC 110, IC 111 for BTech, Ordinary Differential Equations for M.Sc/MTech/PhD
Intended for	Elective for M.Sc./ MTech/PhD/BTech (All Branches)
Mutual Exclusion	None
Approval	46th BoA (May 2022)

Course Contents

- **Module 1:** Introduction to Nonlinear Dynamics and Chaos, Recent applications of Chaos, Computer and Chaos, Dynamical view of the world (3hours)
- **Module 2:** Basics of nonlinear science: Dynamics, Representations of Dynamical Systems, Types of Dynamical Systems, Nonlinearity, Vector Fields of Nonlinear Systems, Nonlinear systems and their classification, Dissipative Systems, Deterministic vs. Stochastic Systems, Degree of Freedom, State Space, Phase Space, Attractor (5 hours)
- **Module 3:** Existence and uniqueness of solutions, Fixed points and Linearization, Flows on line, Fixed Points and its Stability, Analytical Approach, Graphical approach, Simulation of Equations (5 hours)
- **Module 4:** Elementary Bifurcation Theory: Saddle Node, Transcritical, Pitchfork, Imperfect, Hopf bifurcation (4 hours)
- **Module 5:** Two dimensional Flows, Simple Harmonic Mass-Spring Oscillator (4 hours)
- **Module 6:** Limit Cycle, Ruling out closed orbits, Poincare Benedixson theorem (4 hours)
- **Module 7:** Chaos and tools for its Detection: Chaos and Butterfly effect (SDIC), Center manifold theory and Poincare maps, Lyapunov Exponents, Power spectrum, phase, Stable and Unstable Manifolds, Frequency Spectra of Orbits, Dynamics on a Torus, analysis of Chaotic Time Series. Examples of chaotic systems: Lorenz Equations, Application of Chaos in sending secret messages, Rossler Equations, Chua's Circuit, Introduction to Fractals, Dimensions of fractals, Cantor Set and Koch curve (6 hours)
- **Module 8:** One dimensional map, Logistic Map, Henon map, Period doubling Route to chaos, Feigenbaum constants (5 hours)
- **Module 9:** Statistical description of Chaotic Systems: The concepts of invariant measure, Sinai-Ruelle-Bowen measures, ergodicity and mixing, Lyapunov exponents, and the dynamical (Kolmogorov-Sinai) entropy, and connecting them to the fractal dimension of invariant sets, and to the escape rate from a chaotic repeller. (6 hours)

Text Books

1. H.G. Solari, M.A. Natiello and G.B. Mindlin, **Nonlinear Dynamics: a two-way trip from Physics to Maths**, Overseas publication, 2019.
2. Jordan, D. W., and P. Smith., **Nonlinear Ordinary Differential Equations**, Oxford University Press 2007

References

1. K. Allgood, T.Sauer, J.A.Yorke, **Chaos: An Introduction to Dynamical systems**, Springer Verlag 1998.
2. Ian Stewart, **Does God Play a Dice? The Mathematics of Chaos**, Blackwell.
3. Laksmanan M Rajsekhar, **Nonlinear Dynamics Integrability Chaos and Pattern**, Springer.
4. F.C. Moon, **Chaotic and Fractal Dynamics**, Wiley
5. M W Hirsch, S Smale, R L Devaney, **Differential Equations, Dynamical Systems, and an Introduction to Chaos**
6. Anatole Katok et Boris Hasselblatt, **Introduction to the modern theory of dynamical systems**, Cambridge University Press, 1995
7. Peter Walters, **An introduction to ergodic theory**, Springer, 1982